

Research Statement

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Interpretability results increasingly function as safety evidence: if an internal representation can be located, steered, or ablated, we conclude that the behaviour it governs is understood — and controllable. My research asks what warrants that conclusion. The question grew out of philosophy of language: my ESSLLI 2025 paper argued that context-dependent semantics offers a substantive explanatory frame for transformer behaviour, and the invited follow-up (*under review, Journal of Logic, Language and Information*) tests that frame causally rather than descriptively. The through-line of my work since is to replace plausibility assumptions with falsifiable preservation tests.

Three results define the programme so far. **First, a sufficiency result:** contextual token states at characteristic depths causally govern a controlled meaning-competence profile (disambiguation, counterfactual sensitivity, discourse coherence) under sham-controlled activation patching. **Second, a limitation result:** geometric fidelity does not certify causal preservation. In Gemma-3-4B, SAE reconstructions at cosine similarity 0.997 preserved only ~62% of the causal effect of the underlying activation, while a sibling layer at matched fidelity preserved it fully — across layers, geometric and causal rankings come apart. **Third, a safety instantiation:** the standard single-direction refusal intervention fails a matched replication on Gemma-3-4B-IT — 0 of 135 candidate directions pass a pre-registered selectivity gate, a bounded-search null consistent with concurrent findings (arXiv:2606.23671) — and on Gemma-2-2B-IT, two SAE layers matched in cosine similarity to within 0.002 preserve 0.34 versus 0.70 of the refusal-ablation effect on held-out HarmBench prompts. Stated scope limits apply throughout: pretrained SAEs on instruction-tuned models, single-model cohorts.

As a PhD student I want to develop this into a **causal-preservation benchmark for white-box interpretability and control methods:** multiple model families and behavioural endpoints; multiple representation bases (SAE reconstructions and features, matched-rank PCA, random orthogonal projections, circuit- and attribution-based representations); pre-registered inclusion gates and matched interventions throughout; and robustness tested under context variation, ambiguity, and distribution shift. The target is the audit gap my results expose: a control method can look faithful on every standard metric and still fail to preserve the behaviour it claims to govern. That gap is the benign case of a question you have raised in adversarial form — whether mechanistically stealthy behaviour is achievable, where internal patterns resemble the faithful case while the behaviour diverges. My results suggest that the representational-similarity metrics such a claim would rest on already fail without an adversary, which makes them a weak foundation for activation-based monitoring and control.

I run this programme independently end to end: reproducible intervention pipelines (TransformerLens, activation patching, bootstrap inference, deterministic runners with provenance manifests), built and executed on my own infrastructure. My formal background in logic and philosophy of language is not ornamental here — it is what keeps the evidence standards honest.